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**Experiment Title : Demonstration of Router Hardware and
Connecting Different LANs**

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Experiment No. 07: Demonstration of Router Hardware and Connecting Different LANs

Aim

To identify the hardware interfaces of a Router and to design, configure, and verify the connectivity between two distinct Local Area Networks (LANs) using a Router in Cisco Packet Tracer.

Theory

1. Network Router

A **network router** is a networking device that connects multiple networks and forwards data packets between them. It operates at the **network layer (Layer 3)** of the OSI model.

A router uses **IP addresses** to determine the best path for data transmission.

Functions:

- Connects different networks (e.g., LAN to Internet)
- Routes data packets to the correct destination
- Selects the best path using routing algorithms
- Provides security features like NAT and firewall

2. Role of Default Gateway

A **default gateway** is a device (usually a router) that acts as an **access point to other networks**, especially the internet.

Role:

- When a device wants to send data outside its local network, it sends it to the default gateway

- The gateway forwards the data to the appropriate external network

Example:

- If a computer in a LAN wants to access a website, the request goes to the default gateway (router), which then sends it to the internet

Conclusion:

The default gateway acts as a **bridge between a local network and external networks.**

3. Difference between Router and Switch

Feature	Router	Switch
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Address Used	IP Address	MAC Address
Function	Connects different networks	Connects devices within a LAN
Data Handling	Routes packets intelligently	Forwards frames within network
Broadcast	Does not broadcast to all	Can broadcast (in some cases)

Conclusion:

- Router connects **different networks**
- Switch connects **devices within the same network**

4. Routing Table

A **routing table** is a data table stored in a router that contains information about different network paths.

It helps the router decide **where to send data packets**.

Contents of Routing Table:

- Destination network
- Next hop (next router)
- Interface
- Metric (cost of path)

Working:

1. Router receives a data packet
2. It checks the destination IP address
3. Looks up the routing table
4. Selects the best path
5. Forwards the packet to the next hop

Conclusion:

A routing table ensures that data packets are sent through the **most efficient path to reach their destination**.

Hardware Components Required

- **PC-PT (x6):** End devices acting as network nodes (3 for LAN 1, 3 for LAN 2).
- **Switch 2960 (x2):** Two 24-port switches to create independent local networks. One switch for each LAN.
- **Router ISR 4321 (x1):** The device to connect the two LANs.

- **Cabling:** Copper Straight-Through cables for all connections.

Hardware and Connection Setup

- LAN 1 (Network ID): 192.168.1.0 | Subnet Mask: 255.255.255.0
- LAN 2 (Network ID): 10.0.0.0 | Subnet Mask: 255.0.0.0
- PC Naming Convention:
 - LAN 1: S11_01_A, S11_01_B, S11_01_C (IPs: .2, .3, .4)
 - LAN 2: S11_01_D, S11_01_E, S11_01_F (IPs: .2, .3, .4)

Procedure

Step 1: Physical Topology Design

- Place the ISR 4321 Router in the center of the workspace.
- Place two 2960 Switches (one for each LAN) and six PCs (three per switch).
- Naming: Rename PCs following the convention (e.g., S11_01_A through S11_01_F).

Step 2: Hardware Identification

- Click on the Router and select the Physical tab.
- Observe the modular slots and the GigabitEthernet ports.
- Note the power switch; ensure the router is turned ON.



Step 3: Physical Cabling

- Connect Switch 0 to Router interface GigabitEthernet 0/0/0.
- Connect Switch 1 to Router interface GigabitEthernet 0/0/1.
- Connect all PCs to their respective switches.

- Observation: Note that the link lights between the switches and the router are Red. This is because router ports are disabled by default.

Step 4: Router Interface Configuration (The "Bridge" Setup)

1. Click **Router > CLI**. Type 'no' at the initial setup prompt.
2. Enter the following commands to configure the gateways:
 - Router> enable
 - Router# config t
 - Router(config)# interface gigabitEthernet 0/0/0
 - Router(config-if)# ip address 192.168.1.1 255.255.255.0
 - Router(config-if)# no shutdown
 - Router(config-if)# interface gigabitEthernet 0/0/1
 - Router(config-if)# ip address 10.0.0.1 255.0.0.0
 - Router(config-if)# no shutdown

Observation: The link lights should now turn Green.

Command	Purpose / Function
enable	Moves the user from User EXEC Mode (view only) to Privileged EXEC Mode . This allows the user to view detailed configurations and enter setup modes.
config t (configure terminal)	Moves the router into Global Configuration Mode . This is the only mode where you can make changes that affect the router as a whole.
interface gigabitEthernet 0/0/0	Moves the router into Interface Configuration Mode . It tells the router, "The next commands I type are only for this specific physical port."

ip address [IP] [Mask]	Assigns a logical Identity (IP) and defines the network size (Subnet Mask) for that interface. This becomes the Default Gateway for that LAN.
no shutdown	By default, Cisco router ports are "Administratively Down" (turned off for security). This command powers on the interface to allow data flow.

Step 5: IP Addressing (Logical Setup)

- **For LAN 1 PCs:** Set IPs 192.168.1.x and set Default Gateway to 192.168.1.1.
- **For LAN 2 PCs:** Set IPs 10.0.0.x and set Default Gateway to 10.0.0.1.

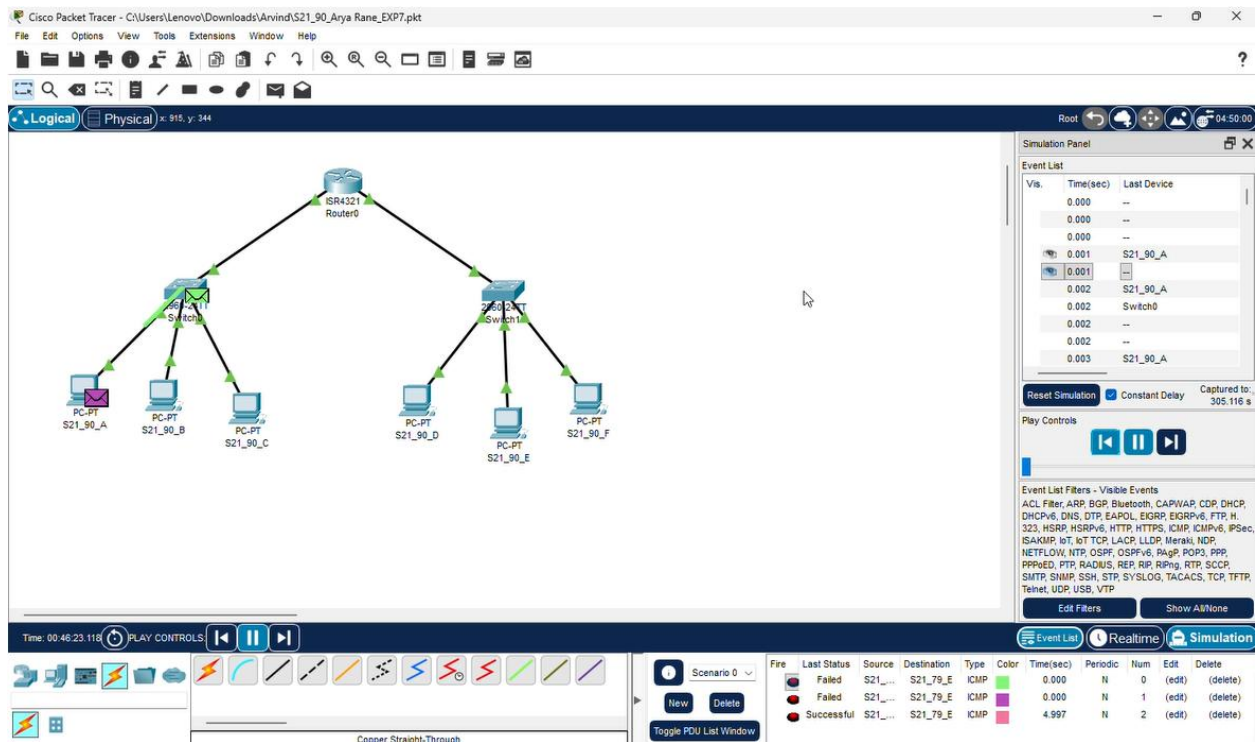
Step 6: Testing and Simulation

1. Real-time:

- Open Command Prompt on **S11_01_A** and type ping 10.0.0.4.
- Verify connectivity. (Note: The first ping may timeout due to ARP).

2. Simulation Mode:

- Filter for ICMP. Use the Add Simple PDU (Envelope) to click S11_01_A (LAN 1) then S11_01_F (LAN 2).
- Observe the packet crossing from the 192.x network to the 10.x network through the router.



Observations

- **Gateway Necessity:** If the "Default Gateway" is not configured on the PCs, the packet never reaches the router and communication fails.
- **Routing Decision:** The Router stops the broadcast from LAN 1 and intelligently forwards the packet specifically to the interface connected to LAN 2.
- **Different Subnets:** Communication was successfully established between two completely different IPs, which is impossible with only a switch.

Conclusion

The experiment successfully demonstrates that a **Router** is the essential hardware for Inter-LAN or Inter-Network communication. It proves that while switches manage local traffic, routers are required to cross network boundaries by using logical IP addresses.